

EVSP DR Current Research

Week of 14 September. Results checked 12 September 2026 at 01:27 EDT; overnight extensions are still running. Open [Figures with explanations](#) for timing, charging costs and geography, or [CG curves and bus schedules](#) for the restored figure library.

Current finding. With set covering and inherited columns, all six chains match the GIRO fleet at $k=3-6$. Chains 3, 5 and 6 also reach ten buses at $k=10$. Larger warm-chain cases remain unfinished.

Integer fleet with inherited columns

Each cell is the number of integer bus routes found. Green matches the target. Red requires extra buses. Grey means no completed result.

Target buses	Chain 1	Chain 2	Chain 3	Chain 4	Chain 5	Chain 6
2	2	2	2	2	3	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	—	7	7	7	7	7
8	—	8	8	8	8	8
9	—	—	9	9	9	9
10	—	—	10	—	10	10

$k=11-15$: no completed warm integer results yet. CG has finished with pricing certificates for chain 3 at $k=11-12$ and chains 5–6 at $k=11$. The overnight extension is queued through $k=15$ for all six chains.

$k=2$ starts from single-trip routes; later sizes reuse validated previous- k columns. Chain 3 $k=10$ also includes routes from a fresh solver solution. Displayed warm fleet optima are proved within their saved pools. Individual route replay passed; duplicate-trip removal and shared charger capacity are not established by this table.

Fresh starts on the same chains

Each size starts independently. Both tables use covering and 240-kWh batteries with 240-kW charging. Fresh fleet proof status varies by cell.

Target buses	Chain 1	Chain 2	Chain 3	Chain 4	Chain 5	Chain 6
5	5	5	5	5	6	5
8	9	9	9	9	9	8
10	11	11	11	11	11	11
15	18	17	18	19	16	19

Inherited columns improve the available integer pool: chain 3 at $k=8$ improves from a proved nine-bus pool optimum to eight; chain 5 at $k=5$ improves from a proved six-bus pool optimum to five.

What is limiting progress

Checking saved routes took too long. Before starting CG on a larger trip set, we check that routes saved from the previous size are still valid. In chain 1 at $k=7$, chain 2 at $k=9$ and chain 4 at $k=10$, this checking used up the available time. We now check at most 512 saved routes, for at most 15 minutes, then let CG search for new routes. All three cases have now finished CG. Their new integer results are still pending. Using fewer saved routes may affect the final integer solution, so this is a separate experiment.

Charger limits made the search for new routes very slow. In one three-bus test, the LP solved in 0.006 seconds, but one search for a new route took 7.15 hours. The run ran out of time before it could prove that no improving route remained. Its 16-bus integer solution uses only the routes found so far; it does not show that 16 buses are necessary. The timing figures explain these two different delays.

Case	Baseline	PARX 60 kW	Station capacity	Both
$k=1$, two duties tested	1 each	1 each	1 each	1 each
$k=2$, 23 trips	2	2	3	3
$k=3$, 35 trips	3	3	16	16

Sixteen is the result in a small generated pool, not a proved physical requirement.

Capacity-constrained schedules passed station sweeps. This pilot has no 65% return-SOC floor and is separate from the six baseline chains.

Running overnight

- 37 warm CG cases continue six chains to $k=15$.
- 40 independent CG cases split one 32-duty, 750-trip parent into four groups of eight, using ten different partitions.
- 10 recombination cases retain the component solution and try routes across groups.

Each case has a dependent one-hour, two-stage MIP on the default partition. At 01:27 EDT, 14 CG cases had pricing certificates: seven warm-chain cases and seven decomposition groups.

New completed integer case	Target buses	Buses found	Fleet minimum in saved pool
Partition 0, group 0	8	8	Proved
Partition 0, group 1	8	8	Proved

Both pass individual-route replay and trip coverage. Duplicate removal and shared charger capacity remain unchecked. There is no completed combined 32-duty solution yet.

Default MIPs: the earlier 75 completed with zero confirmed preemptions. The new batch has two completed and eight running, also with zero confirmed preemptions so far.

Methods and evidence

Each CG route costs $100,000 + \text{electricity} + 5$ per charging start. MIP stage 1 minimizes buses. Stage 2 constrains buses $\leq$ the validated stage-1 incumbent and minimizes electricity plus start fees.

A pricing certificate establishes the LP result in the represented graph at its stated tolerance. An unfinished restricted LP objective is not a full-model lower bound. A MIP proof concerns its saved columns. Fleet target attainment, physical replay and shared-capacity validation are separate checks.

[Experiment register and exact source records](#) · [Overnight experiment settings](#) · [Historical document and figures](#)

Storage maintenance, 12 September: 156 cold files were archived losslessly, freeing 133.74 GB. Active research and V2G data were preserved. [Verification and restore instructions](#).